

Linear programming and windowing based feedrate optimization for spline toolpaths



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ABSTRACT

Optimality and computational efficiency are competing objectives in the design of feedrate optimization algorithms for spline toolpaths. This paper presents a new approach, in which the use of Linear Programming (based on the 'pseudo-jerk' concept) is combined with a novel windowing algorithm to handle long toolpaths. The windowing uses the Principle of Optimality to select suitable connection points between adjacent portions of the feed profile. The result is an algorithm with linear computational complexity, and the ability to process long toolpaths in a parallel manner. Simulations and experimental results confirm negligible compromise in machining cycle time and dynamic accuracy.

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1. Introduction

Competitive multi-axis machining requires new algorithms that achieve the highest possible productivity, by taking advantage of portions in the toolpath where process physics and servo dynamics allow the feedrate to be increased further.

Machining related constraints, like cutting force, torque/power, and deflections, under predefined cutting speeds, can be typically expressed as a linear function of the feedrate [1,2]. For servo drives, actuator velocity, acceleration, and jerk limits are commonly adopted as kinematic constraints, to help preserve the drives' operational life, linear behavior (by avoiding saturations), and dynamic positioning accuracy (by limiting high frequency motion commands with sudden changes in acceleration) [3,4].

Under these constraints, time-optimum feedrate planning for freeform toolpaths is challenging, mainly due to the nonlinearities posed by the axis level jerk equations. In literature, this has been tackled using Sequential Quadratic Programming (SQP) [3,5]. SQP, however, is computationally expensive and its use in real-time is challenging. Thus, heuristic algorithms with iterations [4] and simplifying assumptions have also emerged [6,7]. Some of these trade-off optimality (i.e., cycle time reduction) in favor of simplicity and robustness [7], so that they can be more easily embedded inside CNC kernels, than, for example, SQP methods.

In recent literature, feed optimization has been linearized by approximating the jerk with its upper bound, i.e. 'pseudo-jerk' [8,9]. This enables the use of highly robust and efficient Linear

Programming (LP) algorithms, which provide guaranteed global convergence. In handling long toolpaths, most methods in literature follow sequential solution of the feed profile with each windowed solution affecting its subsequent one [2–4,6,7]. However, significant efficiency and flexibility gains could be achieved, if a feed optimizer would be able to take advantage of emerging parallel computation capabilities.

In this paper, the advantages of LP formulation have been merged with a new windowing algorithm, as shown in Fig. 1. The new approach demonstrates near-linear computational time. Different portions of long toolpaths can also be processed in parallel, with minimal impact on the optimality of the final result.

Henceforth, the LP formulation is briefly reviewed in Section 2. The new windowing algorithm is described in Section 3. Section 4 presents simulation and experimental benchmarks with respect to earlier works. This is followed by the conclusions in Section 5.

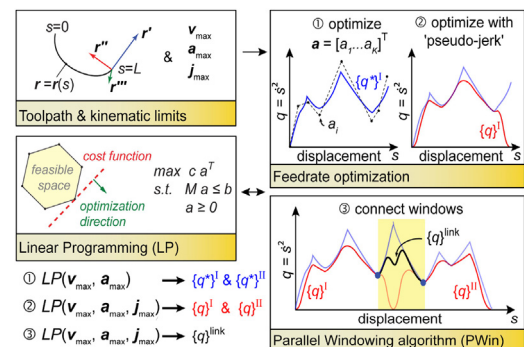


Fig. 1. Overview of the proposed feedrate optimization algorithm.

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2. Linear programming (LP) method for feed optimization

The following presentation considers 3-axes. However the methodology can also be extended to 5-axes, as shown in Ref. [9].

Given a set of discrete points on a toolpath, it is assumed that a C^2 spline (such as cubic B-spline) is already fitted such that the third order parametric derivatives exist [10,11]:

$$\mathbf{r}(s) = [x(s) \ y(s) \ z(s)] \quad \text{with} \quad 0 \leq s \leq L \quad (1)$$

Above, s is the toolpath parameter, typically defined as the spline travel length. Axis level velocity, acceleration and jerk are:

$$\left. \begin{aligned} \mathbf{v} &= \frac{d\mathbf{r}}{dt} = \mathbf{r}'\dot{s} \\ \mathbf{a} &= \frac{d^2\mathbf{r}}{dt^2} = \frac{d}{dt}(\mathbf{r}'\dot{s}) = \mathbf{r}''\dot{s}^2 + \mathbf{r}'\ddot{s} \\ \mathbf{j} &= \frac{d^3\mathbf{r}}{dt^3} = \frac{d}{dt}(\mathbf{r}''\dot{s}^2 + \mathbf{r}'\ddot{s}) = \mathbf{r}'''\dot{s}^2 + 3\mathbf{r}''\dot{s}\ddot{s} + \mathbf{r}'\ddot{\ddot{s}} \end{aligned} \right\} \quad (2)$$

In Eq. (2), $\mathbf{r}'$, $\mathbf{r}''$, and $\mathbf{r}'''$ denote geometric derivatives with respect to s . $\dot{s}$, $\ddot{s}$ and $\ddot{\ddot{s}}$ are parametric velocity (i.e., feedrate), tangential acceleration, and tangential jerk, respectively. The optimization problem that maximizes the feedrate along the toolpath (i.e., which minimizes travel time), subject to limits on the kinematic profiles in Eq. (2) is formulated as follows:

$$\left. \begin{aligned} &\text{maximize} \int_0^L \dot{s} \, ds \quad \text{for } 0 \leq s \leq L \\ &\text{subject to:} \quad \left. \begin{aligned} \mathbf{v}_{\max} &\geq |\mathbf{r}'\dot{s}| \\ \mathbf{a}_{\max} &\geq |\mathbf{r}''\dot{s}^2 + \mathbf{r}'\ddot{s}| \\ \mathbf{j}_{\max} &\geq |\mathbf{r}'''\dot{s}^2 + 3\mathbf{r}''\dot{s}\ddot{s} + \mathbf{r}'\ddot{\ddot{s}}| \end{aligned} \right\} \end{aligned} \right\} \quad (3)$$

Above, $\mathbf{v}_{\max}$, $\mathbf{a}_{\max}$, and $\mathbf{j}_{\max}$ are vector quantities, representing the independent kinematic limits corresponding to each axis.

In Eq. (3), the nonlinear terms $\dot{s}^2$, $\dot{s} \cdot \ddot{s}$, and $\dot{s}^3$ make the problem computationally expensive. Thus, a new feed parameter is introduced (q : corresponding to the *square of feedrate*) to linearize the inequalities in Eq. (3) by the following substitutions:

$$\left. \begin{aligned} \dot{s} &= \sqrt{q}, \quad \dot{s}^2 = q, \quad \dot{s}^3 = q\sqrt{q} \\ \ddot{s} &= \frac{1}{2}q', \quad \dot{s} \cdot \ddot{s} = \frac{1}{2}q'q, \quad \dot{s} \cdot \ddot{\ddot{s}} = \frac{1}{2}q'q' \end{aligned} \right\} \quad (4)$$

Substituting Eq. (4) into Eq. (3), and considering that q is positive by definition, the problem is reformulated as:

$$\left. \begin{aligned} &\text{maximize} \int_0^L q \, ds \quad \text{for } 0 \leq s \leq L \\ &\text{subject to:} \quad \left. \begin{aligned} \mathbf{v}_{\max} &\geq |\mathbf{r}'|\sqrt{q} \\ \mathbf{a}_{\max} &\geq |\mathbf{r}''q + \frac{1}{2}\mathbf{r}'q'| \\ \mathbf{j}_{\max} &\geq |(\mathbf{r}'''q + \frac{3}{2}\mathbf{r}''q' + \frac{1}{2}\mathbf{r}'q'')|\sqrt{q} \end{aligned} \right\} \end{aligned} \right\} \quad (5)$$

The nonlinear terms in Eq. (5) are q' , q'' , and $\sqrt{q}$. This form, however, is more favorable than Eq. (3), since the geometric derivatives q' and q'' can be linearized through adequate parameterization of $q(s)$. The velocity constraint in Eq. (5) is linearized by squaring both sides. In the acceleration and jerk equations, q' and q'' terms are linearized by constructing the profile for q and its geometric derivatives (q' , q'') as a B-spline, using De Boor's algorithm [3,8]:

$$\left. \begin{aligned} q(s) &= \sum_{i=1}^K N_{i,2}(s)a_i = \mathbf{N}(s) \cdot \mathbf{a} \\ q'(s) &= \sum_{i=1}^K N'_{i,2}(s)a_i = \mathbf{N}'(s) \cdot \mathbf{a}, \quad q''(s) = \sum_{i=1}^K N''_{i,2}(s)a_i = \mathbf{N}''(s) \cdot \mathbf{a} \end{aligned} \right\} \quad (6)$$

Above, $N_{i,2}(s)$ is the basis function for a 3rd order B-spline. K is the number of discretization points. $\mathbf{a} = [a_1 \ \dots \ a_K]^T$ is the vector of control points, which essentially become the free variables of the

feed optimization problem (Fig. 1). Thus, the expressions for q , q' , and q'' inside Eq. (5) assume a linear form in terms of $\mathbf{a}$. However, the $\sqrt{q}$ term in the jerk inequality still poses a nonlinearity. To resolve this, q is replaced by a precomputed upper bound q^* (Fig. 1). A suitable choice for q^* is the solution obtained by considering only the velocity and acceleration constraints in Eq. (5), which guarantees a unique solution with a larger feedrate than any other feasible solution [8,9]. Thus, the jerk constraint is replaced with a limit on another variable named 'pseudo-jerk' $\tilde{\mathbf{j}}$:

$$\mathbf{j}_{\max} \geq \tilde{\mathbf{j}} = |(\mathbf{r}'''q + \frac{3}{2}\mathbf{r}''q' + \frac{1}{2}\mathbf{r}'q'')|\sqrt{q^*} \quad (7)$$

It can be verified that since $q^* \geq q$, $|\tilde{\mathbf{j}}| \geq |\mathbf{j}|$ will also hold. In other words, as long as the pseudo-jerk constraint in Eq. (7) is satisfied, the actual jerk in Eq. (5) will be limited by $\mathbf{j}_{\max}$. The replacement of the jerk constraint with its relaxed form does not lead to a significant loss of optimality, as q^* is usually close to q .

Considering the B-spline definition for q , q' , q'' in Eq. (6), and the objective function and inequality constraints in Eqs. (5) and (7), the optimization can be written in compact LP form as in Eq. (8).

$$\text{maximize: } \mathbf{c}^T \mathbf{a}, \quad \text{subject to: } \left. \begin{aligned} \mathbf{M}\mathbf{a} &\leq \mathbf{b} \\ \mathbf{a} &\geq 0 \end{aligned} \right\} \quad (8)$$

Above, $\mathbf{M}$ is a constant matrix determined by substituting the expressions for q , q' , and q'' in Eq. (6) into the constraints in Eqs. (5) and (7). Thus, the terms inside $\mathbf{M}$ depend on the geometric derivatives $\mathbf{r}'$, $\mathbf{r}''$, $\mathbf{r}'''$, and basis functions $N_{i,2}$, $N'_{i,2}$, $N''_{i,2}$ evaluated at the selected K points along the toolpath. Vector $\mathbf{b}$ contains the axis kinematic limits ($\mathbf{v}_{\max}$, $\mathbf{a}_{\max}$, $\mathbf{j}_{\max}$). $\mathbf{c}$ is typically chosen as a uniformly distributed weighting vector, placing equal importance in maximizing every local feedrate control point contained within the variable vector $\mathbf{a}$. Upon determining q^* , by first considering only the velocity and acceleration limits, the full problem also incorporating the pseudo-jerk limit, is solved afterwards. Compared to more elaborate solvers like SQP, LP is much faster and achieves convergence to a global optimum.

3. Parallel Windowing (PWin) algorithm for long toolpaths

LP formulation greatly reduces the computational load. However, in freeform machining, toolpaths can comprise ten thousands of NC position commands. Attempting to solve the feedrate through LP for such long toolpaths in one shot would be highly inefficient, and possibly infeasible, considering the capabilities of current CNC systems. The new PWin algorithm, presented here, was developed to address this problem.

The basic idea behind the windowing is illustrated in Fig. 2. Comparing the time-optimized feed profiles (shown in terms of q) for a complete (global) toolpath and its windowed (local) portion, it can be seen that once the transition zones due to the (zero) boundary condition (BC) are removed, there is a direct overlap between the two profiles in the green shaded region, where solely the toolpath geometry determines the optimized feed profile.

Hence, as long as local toolpaths can be solved for long enough windows containing BC dominant zones and the geometry influenced portion, then it should be possible to reconstruct the optimized feed for the complete (global) toolpath, simply by shifting the window's start and end points, and in each case identifying the region of overlap between the local and global solutions. This follows from the Principle of Optimality [12], which implies that a short section taken out of a long trajectory that is optimal, must itself be optimal as well.

In identifying the overlap region, the local minima in the windowed solution right after the initial and just before the final BC can be used. These minima (i.e., local reductions in feedrate) are caused by the curvature of the toolpath, and not by the BC's themselves. Hence, the same minima also exist in the feed profile optimized for the complete (global) toolpath. Based on the above insight, the following algorithm is proposed (illustrated in Fig. 3):

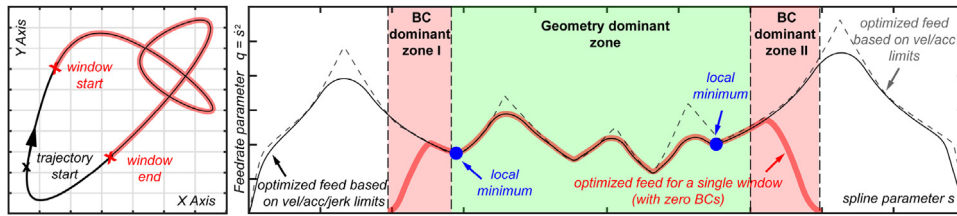


Fig. 2. Optimized feed profile for a long (global) toolpath and its windowed (local) portion, considering zero boundary conditions for both cases.

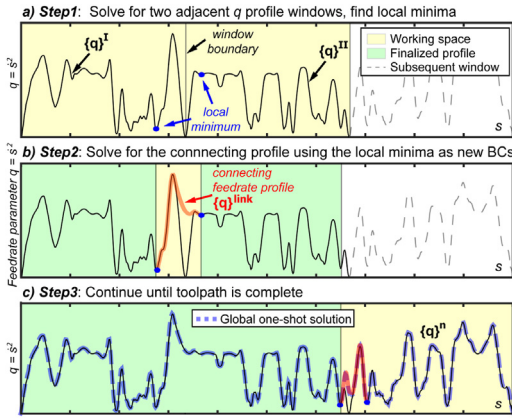


Fig. 3. Progression steps of the windowing algorithm.

Step 0: The toolpath is parsed into individual windows of length L . In order to reach the geometry dominant overlap, shown in Fig. 2, the window size has to be large enough to accommodate the distance (L_0) required to accelerate or decelerate under rest conditions and maximum feedrate. Practically: $L \cong (3 \dots 10)L_0$.

Step 1: Two adjacent windowed portions of the toolpath are solved using the ‘pseudo-jerk’ approximation via LP, considering zero velocity and acceleration BC. These solutions, named $\{q\}^I$ and $\{q\}^{II}$, have guaranteed feasibility due to the choice of BC’s.

Step 2: Blending between earlier solutions $\{q\}^I$ and $\{q\}^{II}$ takes place by utilizing the local minima on each side of the window boundary as connection points. If a local minimum is not available, e.g., due to the toolpath being straight or the window size too small, then the midpoint of the associated window is used. The LP solution, in either case, yields the connection profile $\{q\}^{link}$ which has guaranteed feasibility, due to the precursor solution (i.e., end of $\{q\}^I$ and beginning of $\{q\}^{II}$) already being a feasible candidate for the constraints in Eq. (5).

Step 3: Processing of other adjacent blocks and their blending.

The developed windowing algorithm has the following features:

1. Capability to optimize the feed more efficiently for long toolpaths. The computational time has been benchmarked to have been reduced from $O(K^{4.43})$ to $O(K^{0.98})$, thus leading to near linear-time increase with toolpath size (Fig. 4).
2. Compromise to optimality is minimal, less than 1–2% (Fig. 4). As seen in Fig. 3c, the profile generated with the windowed solution is nearly coincident with that of the one-shot case.
- 3.

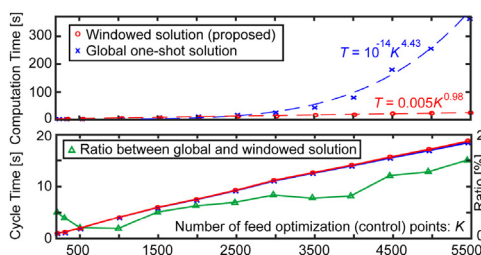


Fig. 4. Comparison between one-shot and windowed solutions.

Implementation of Steps 1 and 2 within themselves is parallelizable, as opposed to the sequential processing used in mainstream feed optimizers. Computations of individual windows and individual connecting profiles are independent.

4. The windowing algorithm can also be paired with feed optimization methods other than LP, such as SQP, VPOp, etc.
5. Even if the window size is kept short, iterating Steps 1 and 2 still produces the optimized feed for the global toolpath. This, however, is not recommended due to computational efficiency considerations.

4. Simulation and experimental results

The proposed LP + PWin algorithm was benchmarked with two other techniques recently reported in literature. One is the Velocity Profile Optimization (VPOp) [4] (which intersects the kinematic constraints at each interpolation step) and the other is the forward projection method [2,7] (which uses the S-curved feed profile as its template). All implementations were realized in Matlab and executed on a Core i5 2.6 GHz processor. The toolpath defined by the waypoints in Fig. 5a was used in the benchmarks. The actuator limits considered were $v_{max} = 150$ mm/s, $a_{max} = 500$ mm/s², and $j_{max} = 10,000$ mm/s³ for the x–y–z axes, belonging to the 2.4×1.2 m² router used in the machining experiments. In implementing the proposed algorithm for the given toolpath, 15 windows and $K = 3000$ constraint checkpoints ($a = [a_1 \dots a_K]^T$) were used. The feedrate profile obtained with the LP + PWin algorithm has been overlaid on top of the toolpath, as shown with the color scale in Fig. 5b. The motion times, commanded actuator profiles, and computational times are reported in Figs. 5, 6, and Table 1. The percentages of cycle time increase in Table 1 are reported with respect to the fastest profile (VPOp). Minor violations of the acceleration and jerk limits in Fig. 6 are attributed to the discrete number of constraint checkpoints used.

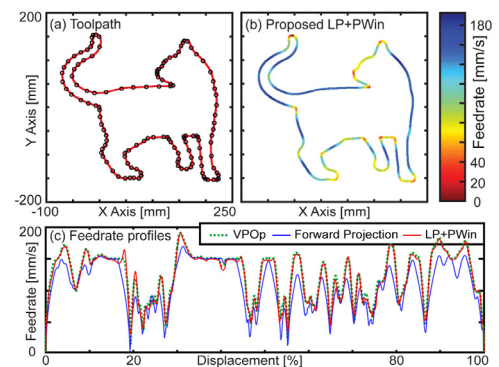


Fig. 5. Test toolpath and feedrate benchmark results. (a) Toolpath, (b) proposed LP + PWin feedrate, (c) feedrate profiles comparison.

Table 1

Comparison of motion (cycle) and computational times.

Feed generation method	Cycle time [s]	Computation time [s]	Cycle time increase
VPOp [4]	15.07	>600	Fastest
Forward projection [2,7]	18.64	3.12	24%
Proposed LP + PWin	15.34	10.39	2%

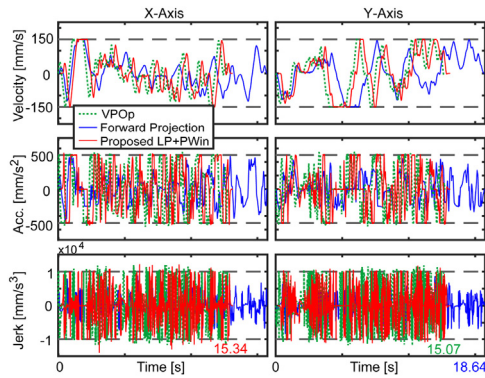


Fig. 6. Kinematic profiles produced by three different feedrate optimization methods.

The benchmarks indicate that the forward projection method, while being the fastest computationally, trades-off performance (i.e., cycle time) by around 24% compared to the other two algorithms. VPOp and the proposed LP + PWin algorithms yield the shortest cycle times and also very similar kinematic profiles (Fig. 5c), although algorithmically the two approaches are very different. The computational time required for LP + PWin appears to be 60 times less than the one for VPOp. The required computational time for LP + PWin is also around 30% less than the optimized motion time, implying the possibility of implementing this algorithm in future real-time applications, as the convergence and feasibility guarantees were already discussed in Section 3.

To further validate the computational robustness in an experimental case, a 3D 6899 mm long toolpath comprising of 1000 waypoints has been used in machining experiments (Fig. 7). In the implementation of LP + PWin, 100 windows and 10,000 constraint evaluation points were used. For comparison, the forward projection method was also implemented. Fig. 8 shows the

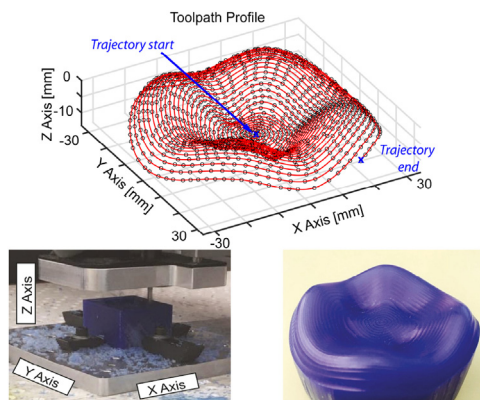


Fig. 7. 3D toolpath, machining setup and finished part.

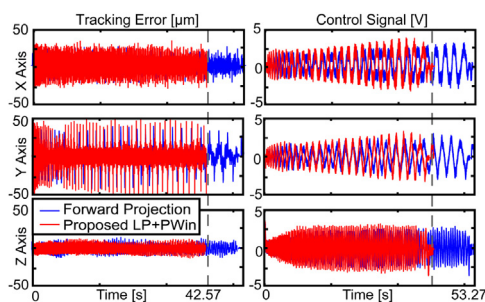


Fig. 8. Tracking error and control signal comparison.

experimental servo tracking errors and control signals obtained with each method. As can be seen, in spite of the 24% cycle time reduction over the forward projection technique, the LP + PWin algorithm is able to keep the servo errors at very similar magnitude. The control signal, on the other hand, is a little larger. This is due to some of the conservative assumptions made by the forward projection algorithm to achieve simplifications that speed up the computation of the feed profile. Thus, LP + PWin makes better use of the actuators' kinematic capabilities, without violating them. This experiment demonstrates that the proposed algorithm can indeed be used to interpolate very long toolpaths.

5. Conclusions

This paper has presented a new feed optimization algorithm, which merges LP based formulation with a novel Parallel Windowing technique (PWin) constructed from the Principle of Optimality. The algorithm has feasibility guarantee. It is robust and demonstrates only linear increase in computational time when handling long and continuous toolpaths, as validated in benchmarks. Opposed to classical feed optimization, which take a sequential approach, the proposed algorithm can solve any number of adjacent windows and their linking trajectories in a parallel manner, thus allowing implementation on parallel computing architectures to support next generation process planning and CNC systems. Such parallelization can also benefit future super high-speed machine tools, by keeping the computational time of very fast trajectories below their actual execution time. Further experiments to validate the algorithm in multi-axis machining of gas turbine components are underway.

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